

eSim Semester Long Internship - Autumn 2026

Task 7 Submission

<https://esim.fossee.in/research-migration-project>

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Title of the circuit : Analog Lock-in Amplifier for Weak Signal Detection

Theory/Description : Lock-in amplifiers utilize the phase-sensitive detection (PSD) technique to isolate a data signal at a specific reference frequency. In PSD-based systems, noise signals at frequencies other than the reference frequency are rejected, meaning they do not significantly affect the desired signal information. The PSD technique is fundamentally based on the synchronous rectification of the data signal, governed by a control signal that determines the amplification sign. When both the data and control signals share the exact same frequency and phase values, the data signal becomes full-wave rectified at the output. Subsequently, a low-pass filter is employed to extract the average value of this processed signal, thereby cancelling the noise and recovering the desired signal information.

Because this fundamental technique requires a perfect match in phase, phase dependency (which often occurs with capacitive sensors) can be eliminated by employing a dual-phase lock-in architecture. This dual-phase architecture adds a second lock-in amplifier branch controlled by a reference signal that is shifted by 90 degrees. To enable single-supply operation (e.g., 3V for battery-operated embedded systems), the system utilizes a quadrature oscillator to provide the 90-degree phase-shifted control signals, and a DC offset is added to center the data signals within the supply range before mixing and low-pass filtering.

Reason to reproduce with eSim : Reproducing this circuit using eSim is highly valuable as it demonstrates the design of a complete, dual-phase analog lock-in amplifier optimized for a single 3V supply, which is critical for battery-operated wireless sensing networks. Simulating this system in an open-source environment provides immense educational value by allowing users to verify how phase-sensitive detection can successfully recover small AC sensor signals from extreme noise environments with a signal-to-noise ratio below 0.025. Furthermore, eSim

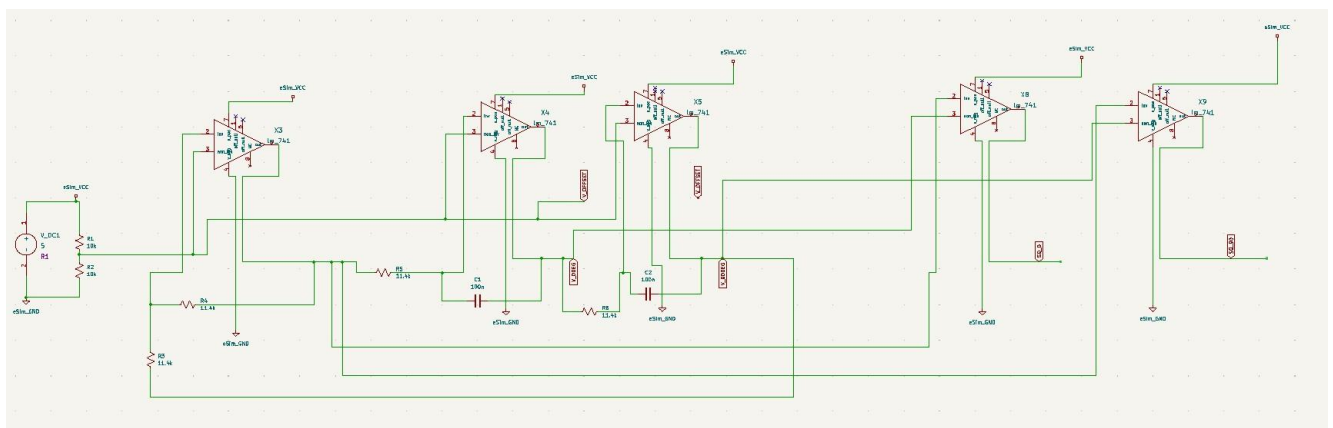
is perfectly suited to simulate and verify the circuit's specific building blocks, such as the quadrature oscillator that eliminates phase dependency and the digitally switched analog mixers that are modified to handle 1.5V DC-offset signals.

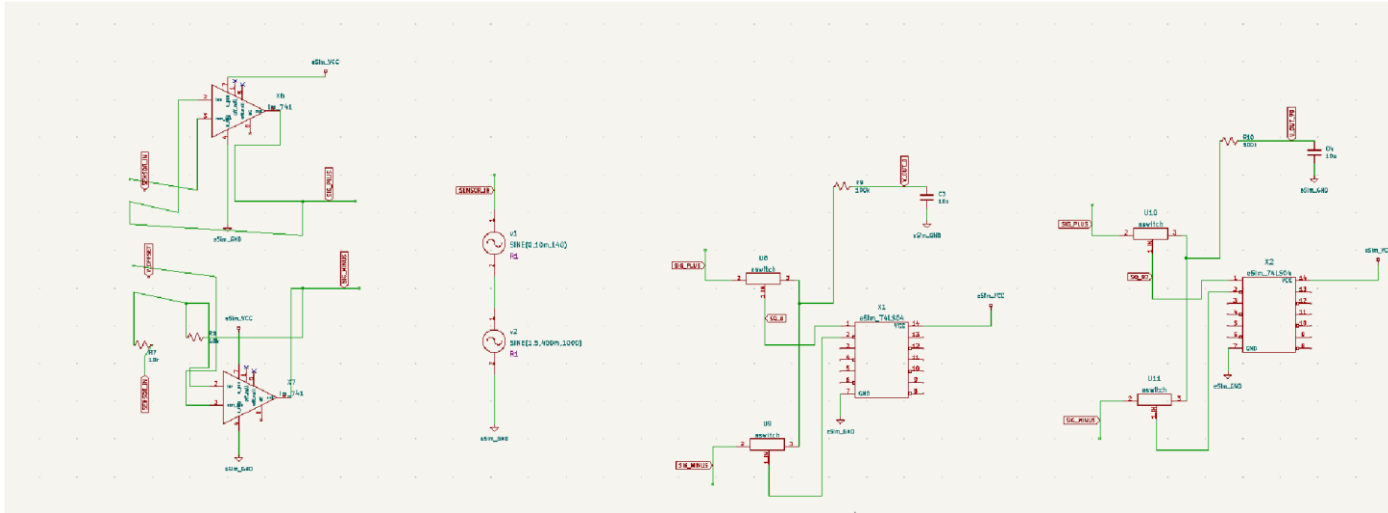
Expected Outcome/outputs : The simulation is expected to demonstrate the successful recovery of a small AC signal that is heavily buried in noise, regardless of phase dependency. The measurable outputs of the circuit will be two distinct DC signals (V01 and V02) that represent the low-pass filtered, synchronously rectified outputs from the dual-phase branches, both centered around a 1.5V DC offset. The circuit's performance will be validated by applying the extracted DC voltages to the mathematical recovery formula:

$$A = \frac{\pi}{2} \sqrt{(V_{01} - 1.5V)^2 + (V_{02} - 1.5V)^2}$$

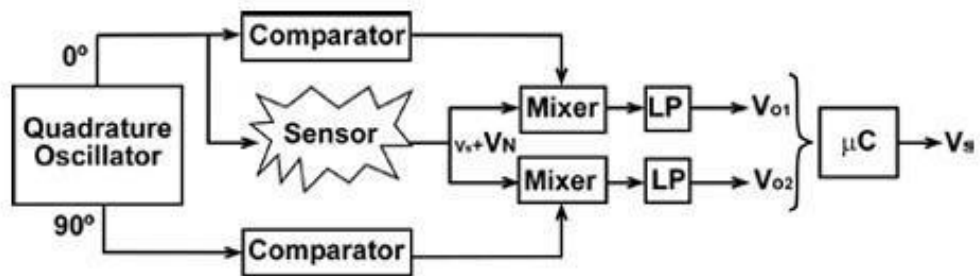
which calculates the noise-free amplitude (\$A\$) of the original sensor signal. Ultimately, the simulation should verify the system's ability to accurately recover signal information even when the Signal-to-Noise Ratio (SNR) is below 0.025 (meaning the noise amplitude is up to 40 times higher than the data signal), with a measurement error of less than 9%.

Circuit Diagram(s) :

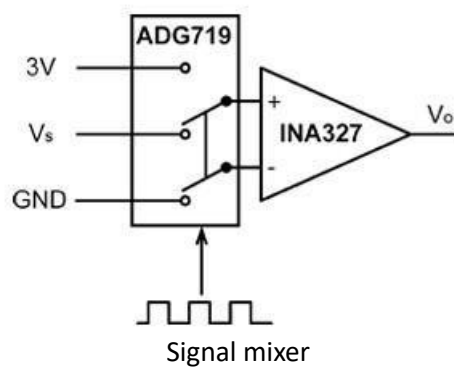




Block Diagram (s) :



Block scheme of the proposed lock-in system



- An input consisting of target sensor data plus noise is centered at a 1.5V DC offset and fed into two parallel signal mixers.
- A local quadrature oscillator generates two 140 Hz reference sine waves shifted by 90 degrees.
- Comparators convert these reference sine waves into square waves to control the switching of the respective signal mixers.

Inputs:

- An AC sensor data signal heavily masked by noise (capable of reaching a Signal-to-Noise Ratio of 0.025), centered around a 1.5V DC offset.
- Two 140 Hz reference sine waves with a 90-degree phase shift and a 1.5V DC offset level, generated internally by the quadrature oscillator.

Intermediate Waveforms:

- Square wave control signals produced by the comparators from the oscillator's sine waves.
- Synchronously rectified transient waveforms at the output of the two signal mixers, which invert the signal around the 1.5V axis depending on the square wave control state.

Final Outputs:

- Two stable DC voltages (V01 and V02) measured at the output of the 1 Hz low-pass filters.
- These DC voltage readings will contain the 1.5V offset and the extracted average values, which can be mathematically combined to verify the noise-free amplitude of the original sensor signal with an estimation error below 9%.

Footprint Assignment Details:

- **Operational Amplifiers (LM741):** Components X3, X4, X5, X6, X7, X8, and X9 are assigned to standard 8-pin Through-Hole Dual In-line Packages (Package_DIP:DIP-8_W7.62mm).
- **Logic Inverters (74LS04):** The hex inverter chips X1, X2, and X10 are assigned to 14-pin Through-Hole Dual In-line Packages (Package_DIP:DIP-14_W7.62mm).
- **Resistors (10k, 11.4k, 100k):** All resistors (R1 through R10) are assigned to standard axial through-hole footprints with a 10.16mm pitch (Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal).
- **Capacitors (100nF, 10uF, etc.):** All capacitors (C1 through C14) utilize radial disc through-hole footprints with a 5.00mm pitch and 5.0mm body diameter (Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm).
- **Connectors and Test Points:** The 2-pin external connections (J1, J2, J3, J6) are assigned to standard vertical 1x02 pin headers with a 2.54mm (100 mil) pitch

(Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical). The single-pin test nodes (J4, J5) are assigned to vertical 1x01 pin headers with a 2.54mm pitch (Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical).

Routing Methodology

- **Dual-Layer Stack-Up:** The PCB was routed using a standard two-layer configuration, utilizing both the Front Copper (F.Cu) and Bottom Copper (B.Cu) layers to manage the high density of through-hole intersections.
- **Perimeter "Ring Road" Strategy:** To accommodate the dense central cluster of operational amplifiers and logic ICs, a perimeter routing strategy was employed. Long trace runs, including complex cross-board signal nets, were deliberately steered through the empty canvas around the outer edges of the components rather than forcing them through tight inner component pads. This prevented congestion and eliminated short-circuit risks.
- **Layer Alternation and Vias:** Trace directions were managed by actively switching between the top and bottom copper layers to cross over perpendicular tracks without intersecting. Vias were utilized efficiently to transition between layers, and redundant or stacked vias were meticulously audited and removed to ensure a clean drill file.
- **Power and Ground Distribution:** The VCC and GND nets were carefully routed to every active component. Clearances between power traces and signal pads were strictly maintained to prevent zero-clearance short circuits.
- **Physical Clearances (Courtyards):** Component placement was iteratively adjusted to ensure no overlapping physical courtyards, guaranteeing that the final physical components (such as radial capacitors and DIP sockets) will not collide during the soldering and assembly process.
- **DRC Validation:** The routing was continuously checked against standard manufacturing constraints using the KiCad Design Rules Checker (DRC), resulting in a finalized layout with exactly 0 DRC violations, 0 unconnected pads, and a properly defined Edge.Cuts board outline.

DRC Verification

The final printed circuit board layout was subjected to a comprehensive Design Rule Check (DRC) to ensure strict adherence to electrical and physical manufacturing constraints. Throughout the routing process, clearance violations, overlapping component courtyards, and dangling tracks were systematically identified and resolved.

The final DRC scan validates the integrity of the design, returning the following perfect results:

- **0 DRC Violations:** Confirming that all track clearances, via dimensions, and board outlines meet required fabrication standards.
- **0 Unconnected Pads:** Verifying that 100% of the schematic netlist has been successfully routed with continuous copper tracks.

- **0 Footprint Errors:** Ensuring that all 3D packages and 2D footprint geometries are correctly mapped to the schematic symbols

Design Considerations and Conclusions

Design Considerations: The layout of the Analog Lock-in Amplifier for Weak Signal Detection required careful spatial planning to balance a high component density with a strict two-layer PCB constraint. Standard Through-Hole Technology (THT) components—including DIP packages for the operational amplifiers and logic inverters—were deliberately selected to ensure reliable physical prototyping and ease of manual assembly. A primary design challenge was managing physical footprint clearances (courtyards) while routing intersecting signal paths between the densely packed ICs. By employing a perimeter routing strategy and carefully alternating between the front and bottom copper layers, signal paths were successfully connected without relying on excessive or overlapping vias, thereby reducing manufacturing complexity.

Conclusion: This project successfully demonstrates the complete PCB design workflow using the open-source eSim EDA environment. The complex analog architecture was transitioned from initial schematic capture to a fully routed, manufacturing-ready physical layout. The rigorous application of design rules resulted in a mathematically perfect board, verified by a final Design Rule Check (DRC) yielding 0 violations, 0 unconnected pads, and 0 footprint errors. All necessary fabrication outputs, including Gerber and Excellon Drill files, were successfully generated. Ultimately, this submission proves both the viability of the dual-phase lock-in amplifier circuit for weak signal recovery and the robust capability of eSim to handle advanced analog instrumentation design.

Research Paper/Journal/etc. :

Title : A complete low voltage analog lock-in amplifier to recover sensor signals buried in noise for embedded applications

Author : M. Gabal*, N. Medrano, B. Calvo, P. A. Martínez, S. Celma, M. R. Valero

Page No. : 74-77

Link : [\[PDF\] A complete low voltage analog lock-in amplifier to recover sensor signals buried in noise for embedded applications | Semantic Scholar](#)

Source/Reference(s) :

- Stanford Research Systems, "Lock-in amplifiers, Appl Notes", Data Sheets, 1999. (*This corresponds to the Application Note #3 we discussed earlier, which provides excellent theory on phase-sensitive detection*).
- Meade, M. L., "Lock-in Amplifiers: Principles and Applications", Peter Peregrinus Ltd, 1983.
- National Instruments, "NI USB-621x User Manual", 2009. (*Referenced for the DAQ board used to simulate the low SNR sensor environment*).
- Data sheets for the specific commercial ICs used in the experimental setup: Analog Devices ADG719 (SPDT switch) and Texas Instruments INA327 (Instrumentation Amplifier).

Note: Fields marked with an asterisk (*) are mandatory and must be filled for successful submission.